The 2025 American Association for Thoracic Surgery (AATS) expert consensus document: Surgical management of esophageal and gastroesophageal junction cancer

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ABSTRACT

Objective: Esophagectomy has been a cornerstone of the treatment of nonmetastatic esophageal cancer, but its use has evolved with advances in endoscopic resection techniques, systemic therapy, and radiation. This document reviews the existing literature and reports multidisciplinary consensus and specific recommendations on how best to treat esophageal and gastroesophageal junction cancer at different stages, including difficult clinical scenarios.

Methods: The American Association for Thoracic Surgery Clinical Practice Standards Committee assembled an international, multidisciplinary panel of gastroenterologists, medical oncologists, radiation oncologists, surgical oncologists, and thoracic surgeons with established esophageal and gastroesophageal junction cancer management expertise. A focused literature review was performed with the assistance of a medical librarian. The panel used a modified Delphi method to develop expert consensus statements with a class of recommendations and level of evidence for 3 topics: early-stage adenocarcinoma, locally advanced adenocarcinoma, and squamous cell carcinoma.

Results: Consensus was reached on 23 recommendations, reflecting insights on the basis of the latest literature and current clinical experience that focused on disease stage and histology. The recommendations provide input on the role of surgery, endoscopic therapy, and surveillance for early-stage adenocarcinoma, how to best combine surgery with multimodality therapy for locally advanced adenocarcinoma, and when surgery should be used for patients with squamous cell carcinoma.

Conclusions: Despite the complex nature of esophageal cancer management, consensus on several key recommendations was achieved by this American Association for Thoracic Surgery expert panel. These recommendations, determined on the basis of evidence and developed using the modified Delphi method and expert opinion, provide guidance for thoracic surgeons and other medical professionals who care for esophageal cancer patients. (J Thorac Cardiovasc Surg 2025;170:884-901)

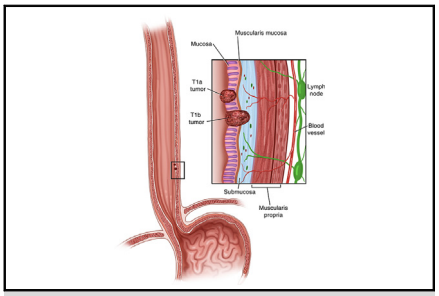

Illustration of the esophagus depicting an intramucosal (T1a) and submucosal (T1b) tumor.

CENTRAL MESSAGE

Decisions regarding the timing and use of esophagectomy, radiation, and systemic therapy for esophageal and gastroesophageal junction cancer must be tailored on the basis of stage and tumor biology.

PERSPECTIVE

The treatment landscape for esophageal cancer is becoming increasingly complex. Advances in systemic therapy coupled with improvements in surgical care—such as minimally invasive techniques and advancements in endoscopic resection—have transformed management. Given this complexity, thorough multidisciplinary review is essential to ensure high-quality care for patients with esophageal cancer.

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Abbreviations and Acronyms

AATS = American Association for Thoracic Surgery
cCR = clinical complete response
CI = confidence interval
CoR = class of recommendation
CROSS = Chemoradiotherapy for Oesophageal Cancer Followed by Surgery Study
CT = computed tomography
ctDNA = circulating tumor DNA
dCRT = definitive chemoradiotherapy
EMR = endoscopic mucosal resection
EUS = endoscopic ultrasound
FLOT = docetaxel, oxaliplatin, leucovorin, and fluorouracil
HR = hazard ratio
ICI = immune checkpoint inhibition
LD = limited data
LoE = level of evidence
LVI = lymphovascular invasion
MAGIC = Medical Research Council Adjuvant Gastric Infusional Chemotherapy
MSI-H = microsatellite instability-high
nCRT = neoadjuvant chemoradiotherapy
NR = nonrandomized
PET = positron emission tomography
SCC = squamous cell carcinoma

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Esophagectomy has been a cornerstone of standard-of-care therapy for resectable esophageal tumors. However, the management of esophageal cancer is complex and directed by several factors, including stage, tumor location, histology, surgical fitness, and a patient's ability to tolerate therapy.[1] Five-year survival rates for patients with esophageal cancer have been historically poor, at less than 5% over the past 50 years. However, this number has improved recently to approximately 21.6% in the United States.[2] This improved prognosis is likely multifactorial. For example, the surveillance of Barrett esophagus has changed stage distribution such that the diagnoses of early-stage cancers are more common.[3] Induction therapy, immunotherapy, targeted therapy, and improvements in the perioperative morbidity associated with esophagectomy have revolutionized therapy for locally advanced cancers.[4,5] Low-risk endoscopic therapies have improved our diagnostic accuracy and reduced the morbidity of curative treatments for early-stage cancers. Lastly, the recognition that the optimal therapy of adenocarcinoma may differ from squamous cell carcinoma (SCC) has improved care for both cancers.[6-10] All of this progress in esophageal cancer management has made the decision process regarding if and when to use esophagectomy more nuanced, particularly for patients with early-stage adenocarcinoma, locally advanced adenocarcinoma, or SCC. This American Association for Thoracic Surgery (AATS) expert consensus document uses a multidisciplinary group of esophageal cancer clinicians to provide recommendations on the use of esophagectomy in these clinical scenarios (Table 1).

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METHODS

This expert consensus process was conducted in line with the joint AATS and Society of Thoracic Surgeons statement on producing clinical practice guidelines (Figure E1).[11] The AATS Clinical Practice Standards Committee initiated this Expert Consensus Document on the Management of Resectable Esophageal and Gastroesophageal Junction Cancer and assembled this writing group of experts. All authors contributed to the drafting of this document and have completed conflict of interest disclosures (Online Data Supplement). This multidisciplinary committee consists of 9 thoracic surgeons, 2 medical oncologists, 2 radiation oncologists, 1 surgical oncologist, and 2 gastroenterologists representing 13 different institutions across 3 continents.

Formulation of Clinical Questions, Literature Review, and Evidence Synthesis

After the creation of the writing group, a literature review was conducted with the aid of librarians trained in systematic review methodology. The search terms and strategy are detailed in Appendix E1. We developed searches that incorporated key words and Medical Subject Headings relating to the following concepts: esophageal cancer, esophagectomy, early-stage disease, resection, recurrence, and complications. Searches were limited to human studies published in English since January 2003.

Themes and concepts were prioritized on the basis of clinical relevance and ambiguity, and an organizational framework was created. This framework included the development of 3 subgroups: (1) early-stage adenocarcinoma, (2) locally advanced adenocarcinoma, and (3) SCC. Each subgroup met several times individually in order to create relevant

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TABLE 1. Complete record of all statements by section

Statement	Section	Class of recommendation	Level of evidence	% consensus
	Esophagectomy for early-stage disease			
1.1	We recommend no further surgical treatment for T1a adenocarcinoma following endoscopic resection when the tumor is well/moderately differentiated, has negative deep/vertical margins, and there are no signs of lymphovascular invasion, given the minimal risk of lymph node metastasis. We also recommend endoscopic treatment of any dysplasia and continued surveillance.	I	B-NR	100%
1.2	Endoscopic appearance of potentially superficial lesions, including endoscopic ultrasound assessment, is not a reliable indicator of depth of disease. Therefore, when feasible, we recommend endoscopic resection with endoscopic mucosal resection/endoscopic submucosal dissection as a diagnostic staging maneuver for suspected T1b and T2N0 patients before deciding on further definitive therapy.	I	C-LD	100%
1.3	We recommend ablation therapy of residual Barrett esophagus after definitive endoscopic treatment of early-stage esophageal adenocarcinoma.	I	B-R	100%
1.4	For T1b tumors and in cases in which there is doubt on margin involvement, it is reasonable to repeat assessment by a dedicated gastrointestinal pathologist to assess whether further treatment is required.	IIa	C-EO	86%
1.5	Endoscopic resection is reasonable alone for low-risk pT1b tumor without lymphovascular invasion that is well to moderately differentiated with superficial extension into the submucosa and negative deep (vertical) margins. Proceeding with esophagectomy after endoscopic resection should be determined on the basis of shared decision-making with the patient on the basis of assessment of risk of lymph node involvement.	IIa	B-NR	100%
1.6	Esophagectomy is recommended for patients who are surgical candidates with T1b tumors with any of the following high-risk features—positive deep (vertical) resection margins, depth of penetration $\geq$500 μm, lymphovascular invasion, poor differentiation—or if the patient is unable or unwilling to undergo intensive surveillance.	I	A	100%
1.7	For patients with T1b adenocarcinoma who undergo endoscopic resection only, we suggest maintaining a strict follow-up schedule to detect any recurrent disease at a curable stage, by endoscopy $\pm$ endoscopic ultrasound every 3 mo for the first year, every 6 mo during the second year, and a minimum of annually thereafter. CT (prefer PET) is advised every 6 mo for 2 y and then annually for 5 total years.	IIa	C-LD	93%
1.8	For patients with cT2N0 esophageal adenocarcinoma, esophagectomy is recommended. The use of neoadjuvant therapy should be based on shared decision-making with the patient on the basis of the risk of lymph node involvement.	I	B-NR	87%
	Esophagectomy for locally advanced disease			
2.1	Diagnostic laparoscopy, including direct visualization, washings, and cytology, should be strongly considered at diagnosis for patients who have signet cell histology or extension at least 2 cm beyond the gastroesophageal junction with at least T3 disease.	I	C-LD	92%
2.2	Perioperative chemotherapy is appropriate and may be preferred over chemoradiation for locally advanced esophageal adenocarcinoma.	I	B-R	94%
2.3	Definitive chemoradiotherapy for locally advanced esophageal adenocarcinoma is reasonable to consider in patients who are not surgical candidates.	IIa	B-NR	100%
2.4	For patients deemed to have clinical complete response after neoadjuvant chemoradiotherapy, it is reasonable to proceed with esophagectomy, as radiographic and endoscopic surveillance has inadequate sensitivity to reliably identify persistent/recurrent disease.	IIa	B-R	77%
2.5	If surgery is deferred after chemoradiation for esophageal adenocarcinoma, surveillance should be done by endoscopy $\pm$ endoscopic ultrasound, as well as imaging (PET-CT strongly preferred), every 3-4 mo for at least the first 2 y and a minimum of annually thereafter for a total of 5 y.	I	C-LD	85%

(Continued)

TABLE 1. Continued

Statement	Section	Class of recommendation	Level of evidence	% consensus
2.6	We recommend testing all patients for mismatch repair at diagnosis.	I	B-R	93%
2.7	For mismatch repair-deficient patients, we advise discussion of upfront immune checkpoint inhibitor and/or surgery.	IIa	B-NR	87%
2.8	For patients with extensive nodal disease and/or "nonregional" lymph nodes, upfront systemic therapy may be preferred.	Ilb	C-LD	86%
2.9	Patients with oligometastatic disease at diagnosis should undergo systemic therapy before being considered for locoregional therapy.	IIa	B-NR	100%
	Squamous cell carcinoma			
3.1	For patients with squamous cell carcinoma, definitive endoscopic therapy is recommended only for those patients with superficial lesions with a low risk of lymph node involvement.	I	B-NR	100%
3.2	For patients with high-risk squamous cell carcinoma who undergo definitive endoscopic therapy only, we suggest maintaining a strict follow-up schedule to detect any recurrent disease at a curable stage, by endoscopy ± endoscopic ultrasound every 3 mo for the first year, every 6 mo during the second year, and a minimum of annually thereafter. CT (prefer PET) is advised every 6 mo for 2 y and then annually for 5 total years.	IIa	C-LD	94%
3.3	Surgery should be considered in patients with esophageal squamous cell carcinoma with a high risk of tumor recurrence despite a clinical complete response to chemoradiation.	I	B-NR	79%
3.4	In patients with recurrent or persistent esophageal squamous cell carcinoma after definitive chemoradiotherapy, salvage esophagectomy is recommended, but it should be performed at experienced centers due to the greater risk of complications.	I	B-NR	100%
3.5	If surgery is deferred after chemoradiation for esophageal squamous cell carcinoma, surveillance should be done by endoscopy ± endoscopic ultrasound, as well as imaging (PET-CT strongly preferred), every 3-4 mo for at least the first 2 y and a minimum of annually thereafter for a total of 5 y.	I	C-EO	93%
3.6	Squamous cell carcinoma of the cervical esophagus should be treated with chemotherapy and radiation, with surgery reserved only in select patients.	I	B-NR	86%

NR, Nonrandomized; *LD*, limited data; *R*, randomized; *EO*, expert opinion; *CT*, computed tomography; *PET*, positron emission tomography.

questions according to the PICO format (population/patient/problem, intervention, comparison, and outcome).

Literature searches were conducted on a regular basis between December 2023 and September 2024. A search of gray literature was performed and supplemented by conference proceedings put forth by the members of this writing group. Members of each subgroup screened the titles and abstracts of the search results for relevance and extracted and synthesized the information from these selected articles.

Evidence-based recommendations were then formulated within subgroups throughout multiple rounds of writing and revisions. The quality of evidence and level of recommendation for each statement were assigned by each subgroup according to language put forth by the AATS and the Grading of Recommendations Assessment, Development and Evaluation (ie, GRADE) system.[11,12]

The Delphi Method

The development of this expert consensus document was conducted using the modified Delphi method. Within subgroups, each recommendation was voted on using a survey with a 5-point Likert scale (1 = strongly disagree; 2 = disagree; 3 = neutral; 4 = agree; 5 = strongly agree). We determined a priori that a response rate of at least 80% was required, with a minimum of a 75% agreement rate to define consensus. Several rounds of surveys were conducted within each subgroup, followed by revisions, until consensus was reached on each statement. Discussions regarding the relevant details, evidence, and references for each statement were held throughout this process. The entire working group had the opportunity to comment on each recommendation prior to publication.

Section 1: Early-Stage Adenocarcinoma

1.1. We recommend no further surgical treatment for T1a adenocarcinoma following endoscopic resection when the tumor is well/moderately differentiated, has negative deep/vertical margins, and has no signs of lymphovascular invasion (LVI), given the minimal risk of lymph node metastasis. We also recommend endoscopic treatment of any dysplasia and continued surveillance. Class of recommendation (CoR): I Level of evidence (LoE): B-nonrandomized (NR)

It has been well established that long-term survival after treatment of esophageal adenocarcinoma is highly dependent on the tumor, node, and metastasis stage of the cancer at the time of diagnosis and cure rates are greatest for individuals in whom the disease is limited to T1 disease.[13] Complete endoscopic resection of early esophageal adenocarcinoma limited to the mucosa, lamina propria, and muscularis mucosa (T1a) has been shown to provide comparable short- and long-term outcomes to those managed with radical esophagectomy.[14,15]

Successful endoscopic therapy is contingent upon complete excision and careful assessment of the resected specimen, including

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characterization of the degree of differentiation, depth of tumor invasion, resection margin, size, and the presence of LVI by pathologists experienced in the evaluation of gastrointestinal specimens. A long-term cure is also dependent on the complete eradication of surrounding dysplastic and non-dysplastic metaplastic epithelium (Barrett esophagus) by resection or ablation and on subsequent close monitoring with regularly scheduled endoscopic surveillance every 3 to 6 months for the first 2 years and annually thereafter for a total of 5 years.[16]

1.2. Endoscopic appearance of potentially superficial lesions, including endoscopic ultrasound (EUS) assessment, is not a reliable indicator of depth of disease. Therefore, when feasible, we recommend endoscopic resection with endoscopic mucosal resection (EMR) or endoscopic submucosal dissection as a diagnostic staging maneuver for suspected T1b and T2N0 patients before deciding on further definitive therapy. CoR: I LoE: C-Limited Data (LD)

Staging the depth of invasion of visible lesions in Barrett esophagus suspicious for adenocarcinoma is challenging (Figure 1). Multiple studies have demonstrated that EUS is not reliable when assessing T stage.[17,18] Deciding on a T stage using EUS or computed tomography (CT) may result in overstaging and subsequent overtreatment of patients if endoscopic treatment is being withheld based on suspected T2 invasion.[19]

The most reliable method for accurate assessment of depth of invasion is histologic assessment, which can be achieved by removing a lesion with endoscopic resection. To assess whether a lesion is amenable for endoscopic resection, endoscopists may use the Paris classification. Lesions classified as Paris type 0-II are generally mucosal lesions, Paris type 0-I lesions are mucosal or submucosal lesions, and Paris type 0-III lesions infiltrate into the muscle layer and are therefore not suited for endoscopic resection.[20]

Assessment of lesions for endoscopic resection should be done by endoscopists with experience in endoscopic resection of early adenocarcinoma.[19] Next to adequate endoscopic evaluation of Barrett esophagus and early cancer, assessment of the resection specimen by an experienced pathologist is important.

1.3. We recommend ablation therapy of residual Barrett esophagus after definitive endoscopic treatment of early-stage esophageal adenocarcinoma. CoR: I LoE: B-NR

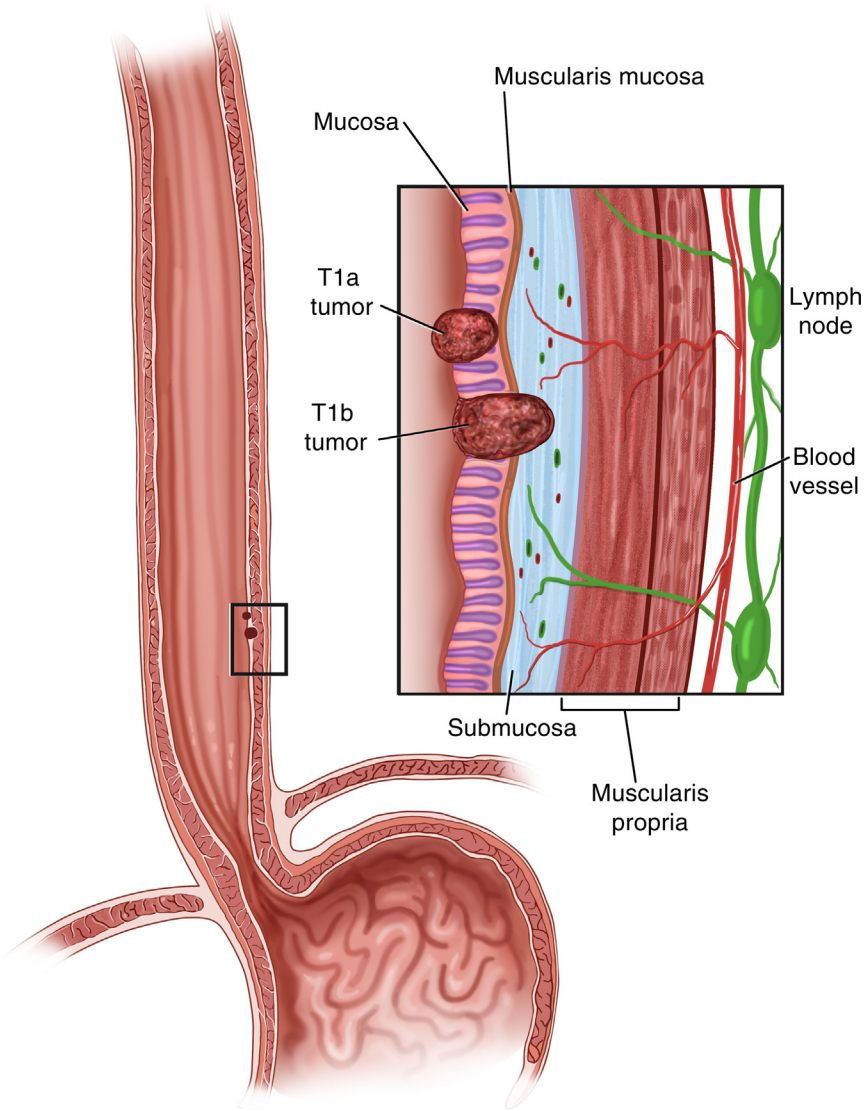

FIGURE 1. Illustration of the esophagus in coronal plane depicting an intramucosal (T1a) and submucosal (T1b) tumor in the midesophagus.

After endoscopic resection of focal lesions with early-stage esophageal adenocarcinoma, there is a 20% to 35% risk of recurrent neoplasia in the remaining Barrett mucosa. To minimize this risk of recurrence, residual flat-type Barrett mucosa can be eradicated using endoscopic ablation.[21-23] A 2-step approach combining endoscopic resection of early adenocarcinoma followed by endoscopic ablation of residual Barrett mucosa has been extensively evaluated in multiple studies.[21,24-28] Complete eradication of all Barrett mucosa can be achieved by stepwise radical endoscopic resection; however, this carries a high risk of esophageal stenosis (88%) and is therefore not advised for flat Barrett mucosa.[24]

Eradication of residual flat Barrett mucosa can also be achieved with argon plasma coagulation, radiofrequency ablation, or cryoablation.[21,25-30] Of these techniques, radiofrequency ablation has been most widely evaluated in multiple high-quality studies, demonstrating high effectiveness with eradication rates >90% and favorable safety profile.[25,26,28] The risks and benefits of ablation of residual Barrett mucosa after endoscopic resection of early-stage adenocarcinoma should be discussed with each patient. For this discussion, one should take into account comorbidity, patient burden, and costs.

In contrast to the American Society for Gastrointestinal Endoscopy recommendations, we advocate for ablation of the cardia after endoscopic resection of a gastroesophageal junction tumor, even when no visible Barrett esophagus is present, because of the increased risk of recurrence at this site. Once ablation is complete and there is no evidence of residual Barrett esophagus, surveillance should follow.[31]

In a case of endoscopic resection of a T1b adenocarcinoma, postponing endoscopic ablation of residual Barrett mucosa might be considered. This may prevent unnecessary treatment in patients who develop early metastasis. In situations of imperfect margins, ablation can bury glands and mask early detection of local recurrence. Furthermore, endoscopic ablation may result in reactive lymph nodes that confound detection of adenopathy by imaging or EUS assessments during the first year of follow-up. Waiting to ablate, however, may lead to quick local recurrence of disease and inability to complete eradication. Typically, ablation is done within 2-3 months, but it can also be done at the same time as resection.

1.4. For T1b tumors, and in cases in which there is doubt on margin involvement, it is reasonable to repeat assessment by a dedicated gastrointestinal pathologist to assess if further treatment is required. CoR: IIa LoE: C-Expert Opinion

T1b esophageal adenocarcinoma is defined by tumor extension beyond the muscularis mucosae and into the submucosa. Critical components of the pathologic assessment of these specimens include determining the depth of tumor invasion, tumor differentiation, and the presence of LVI and assessing both lateral and deep margin status.[32] The presence of these high-risk features substantially increases the risk of recurrence.

Accurate pathologic evaluation of endoscopic resection specimens is critical in order to accurately counsel a patient on the comparative risk of recurrence in surveillance against risk associated with esophagectomy. However, data demonstrate that this is an imperfect process and that when looking at an identical specimen there is significant interoperator variability in pathologic assessment.[33,34] There is evidence to suggest that pathologists should have specific gastrointestinal expertise in order to accurately identify preinvasive from invasive esophageal lesions.[34,35] In addition, there is some evidence to suggest that patients may benefit from referral to a high-volume expert center for treatment of their esophageal cancer.[36] However, this is not possible for many patients. In those cases in which there is a question about margin involvement or classification of a tumor as having high-risk features, it is reasonable to request an evaluation of pathologic specimens by expert gastrointestinal pathologists.

1.5. Endoscopic resection alone is reasonable for low-risk pT1b tumor without LVI that is well to moderately differentiated with superficial extension into the submucosa and negative deep (vertical) margins. Proceeding with esophagectomy after endoscopic resection should be determined on the basis of shared decision-making with the patient on the basis of an assessment of the risk of lymph node involvement. CoR: IIa LoE: B-NR

Endoscopic resection has become the preferred treatment for early-stage/superficial esophageal cancers as a result of improved quality of life and lower risk of complications compared to esophagectomy.[37] The efficacy of endoscopic treatment for T1a tumors is clear, and there is growing evidence that endoscopic therapy is an option for patients with select cT1b tumors with only superficial extension into the submucosa.[15,38] Patients with surgically resected T1b tumors have had greater rates of pathologic lymph node involvement compared with those with T1a tumors. Clinicians must help patients weigh the risks of endoscopic treatment with the possibility of untreated and undetected lymph node disease versus the morbidity and mortality risk of an esophagectomy. Patients with tumors that have more aggressive phenotypes—including LVI, poorly differentiated tumors, or positive deep margins after endoscopic resection, deeper submucosal involvement, or suspected lymph node involvement—may benefit from esophagectomy after endoscopic resection or additional therapy if patients are not surgical candidates.[39-41] There is also evidence to suggest that tumor size >2 cm may add additional risk.[42] Conversely, patients without these features—that is, those with well to moderately differentiated tumors, superficial extension into the submucosa, and negative deep margins—would be considered lower risk within the T1b cohort and could be observed endoscopically. In the absence of these high-risk features, esophagectomy may still be considered in those patients who are unwilling or unable to undergo intensive surveillance detailed elsewhere in this document (statement 1.7).

1.6. Esophagectomy is recommended for patients who are surgical candidates with pT1b tumors with any of the following high-risk features—positive deep (vertical) resection margins, depth of penetration ≥500 μm, LVI, or poor differentiation—or if the patient is unable or unwilling to undergo intensive surveillance. CoR: I LoE: A

There is evidence demonstrating that, compared with low-risk tumors, the risk of lymph node metastasis in patients with clinical T1bN0 tumors with high-risk features such as submucosal invasion beyond 500 μm, LVI, and poor differentiation is significantly increased from around 2% to 50%.[39,42-45] There is also evidence to suggest that tumor size >2 cm may add additional risk.[42] An additional cohort would be those patients with any T1b tumor who are unable to participate in surveillance endoscopy.[46] Esophagectomy should be strongly considered in these patients who have acceptable surgical fitness.[45,47]

1.7. For patients with T1b adenocarcinoma who undergo endoscopic resection only, we suggest maintaining a strict follow-up schedule to detect any recurrent disease at a curable stage by endoscopy ± EUS every 3 months for the first year, every 6 months during the second year, and a minimum of annually thereafter. CT (we prefer positron emission tomography [PET]) is advised every 6 months for 2 years and then annually for 5 total years. CoR: IIa LoE: C-LD

In case it is decided to keep patients under strict follow-up after radical endoscopic resection of a T1b lesion, the aim of follow-up is (1) to detect local intraluminal recurrence at a curable stage, (2) to detect locoregional lymph node metastasis at a curable stage, and (3) to detect distant metastasis with the aim to start early palliative treatment or offer targeted treatment in case of solitary metastasis where appropriate. Locoregional lymph node stations relevant to esophageal cancer are seen in Figure 2.[48]

There are no high-quality data available on the optimal follow-up schedule. The American Society for Gastrointestinal Endoscopy released a clinical practice guideline in 2023 recommending intensive endoscopic surveillance for the

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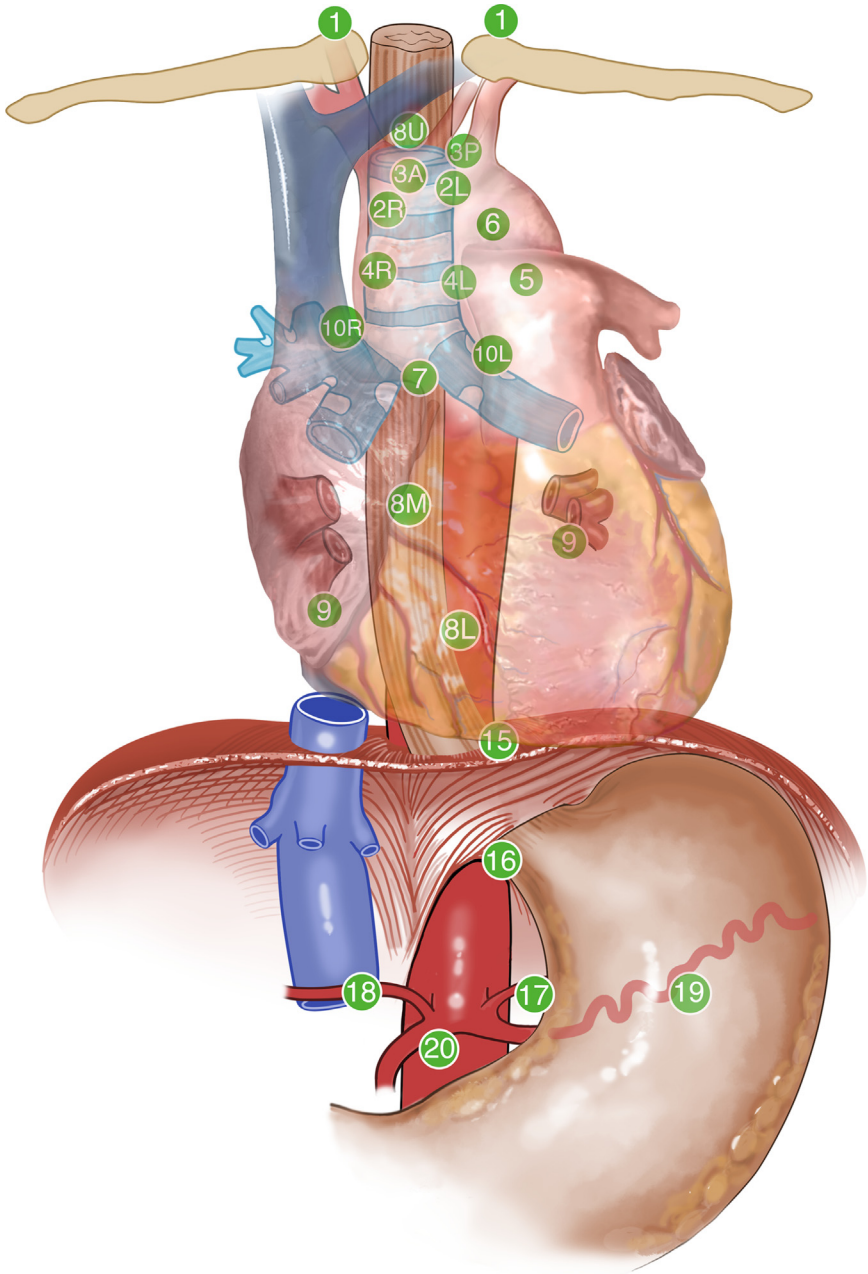

FIGURE 2. Illustration of the lymph node stations of the esophagus as defined by the American Joint Committee on Cancer 8th edition guidelines.[48] Stations 11-14 are intrapulmonary and thus have been excluded from this Figure 1, supraclavicular nodes, above the suprasternal notch; 2R, right upper paratracheal nodes, between the intersection of the caudal margin of the innominate artery with the trachea; 2L, left upper paratracheal nodes, between the top of the aortic arch and apex of the lung; 3P, upper paraesophageal nodes above tracheal bifurcation, located posteriorly; 3A, prevascular, posterior to the sternum; 4R, right lower paratracheal nodes, between the intersection of the innominate artery and trachea and cephalic border of the azygos vein; 4L, left lower paratracheal nodes, between the top of the aortic arch and the carina; 5, aorto-pulmonary nodes, subaortic and paravascular, 6, paraaortic nodes, anterior and lateral to aorta; 7, subcarinal nodes, carina to lower border of bronchus intermedius and L lower lobe bronchus; 8U, upper paraesophageal lymph nodes, from the apex of the lung to the tracheal bifurcation; 8M, middle thoracic paraesophageal lymph nodes, from the tracheal bifurcation to the caudal margin of the inferior pulmonary vein; 8L, lower thoracic paraesophageal lymph nodes, from the caudal margin of the inferior pulmonary vein to the gastro-esophageal junction; 9R, pulmonary ligament nodes, lying in the right inferior pulmonary ligament; 9L, pulmonary ligament nodes, lying in the left inferior pulmonary ligament; 15, diaphragmatic nodes, on the dome of the diaphragm and adjacent to crura; 16, paracardial nodes, immediately adjacent to the gastroesophageal junction; 17, left gastric nodes, along the left gastric artery; 18, common hepatic nodes, on the proximal common hepatic artery; 19, splenic nodes, on the proximal splenic artery; 20, celiac nodes, at the celiac trunk.

first year and annual EUS and CT in this scenario.[49] However, on the basis of available recurrence data and expert opinion, we advise a slightly more aggressive follow-up protocol with endoscopic inspection ± EUS every 3 months for the first year after endoscopic resection, every 6 months during the second year, and a minimum of annually thereafter.[50] Patients with T1b tumors are at particularly high risk of lymph node metastases and recurrence within the first 2 years.[50] Consideration can be given to continued surveillance beyond 5 years depending on patient risk factors.

Dynamic imaging with CT, or preferably PET-CT, is advised every 6 months for 2 years and then annually for a total of 5 years. The aim of CT and PET is mainly to detect lymph node metastasis that may have been missed with EUS and to detect distant metastasis. However, we recognize that this frequency of PET imaging may not be feasible for all patients due to insurance coverage or cost. We reaffirm that many patients with T1b tumors experience distant recurrence, and cross-sectional imaging is an important component of follow-up.

1.8. For patients with cT2N0 esophageal adenocarcinoma, esophagectomy is recommended. The use of neoadjuvant therapy should be determined on the basis of shared decision-making with the patient based on the risk of lymph node involvement. CoR: I LoE: B-NR

Many studies have demonstrated the benefits of multimodal therapy for locally advanced esophageal cancer, including chemotherapy, immune checkpoint inhibition (ICI), radiation therapy, and surgery.[5] Patients with lymph node involvement clearly benefit from a neoadjuvant approach with chemoradiation or systemic therapy followed by surgery.[51] Patients with cT2N0 and cT3N0 tumors are at greater risk of having occult pathologic lymph nodes found after surgery.[52,53] Those who are clinically lymph node negative may not benefit from the addition of neoadjuvant therapy.[54] However, it is recognized that the clinical staging for these tumors is significantly inaccurate.[55] LVI, larger tumor size (>3 cm), high-grade tumors, and signet ring features have all been associated with clinical upstaging after surgery.[56,57] For patients with clinical T2N0 esophageal adenocarcinomas without high-risk features, it is reasonable to proceed with esophagectomy and to treat with adjuvant therapy if indicated on pathologic staging. Neoadjuvant therapy is still an option for patients in this setting and may be of benefit to those patients with cT2N0 tumors with high-risk features.

Section 2: Locally Advanced Adenocarcinoma

2.1. Diagnostic laparoscopy, including direct visualization, washings, and cytology, should be strongly considered at diagnosis for patients who have signet cell histology or extension at least 2 cm beyond the gastroesophageal junction with at least T3 disease. CoR: I LoE: C-LD

Although diagnostic laparoscopy is incorporated into the initial clinical staging algorithms and guidelines for gastric cancers, it is considered optional in the baseline staging evaluation for esophageal cancer patients.[1] Risk factors for peritoneal metastases include cT3 or cT4 tumors, poorly differentiated adenocarcinoma, signet ring cell component, and distal tumor extent of >2 cm beyond the gastroesophageal junction.[58] The predicted risk of peritoneal metastases at 2 cm beyond the gastroesophageal junction is 7.5% and increases up to 35% at 4 cm and beyond with a signet ring cell component.[58] With the 3 risk factors together, patients with cT3-4 tumors, gastric extent ≥4 cm, and signet ring cell component had the greatest rate of peritoneal metastasis (46.7% vs 12% among patients with 1-2 risk factors and 0% with no risk factors; $P = .001$).[59]

Laparoscopic staging with lavage cytology in esophageal adenocarcinoma can aid in accurate staging, especially in patients with identified risk factors.[60] Incorporating subphrenic lavage alongside pelvic lavage during staging laparoscopy significantly enhances sensitivity for detecting peritoneal disease.[61]

2.2. Perioperative chemotherapy is appropriate and may be preferred over chemoradiation for locally advanced esophageal adenocarcinoma. CoR: I LoE: B-R

Both perioperative chemotherapy and preoperative chemoradiation improve survival over primary surgery for locally advanced esophageal adenocarcinoma. The benefits of perioperative chemotherapy (epirubicin, cisplatin, fluorouracil) were initially established by the landmark Medical Research Council Adjuvant Gastric Infusional Chemotherapy (MAGIC) trial, which was composed of patients with gastric adenocarcinoma that included 25% of patients with distal esophagus or gastroesophageal junction cancers.[62] The 2019 FLOT4 trial later showed that perioperative FLOT chemotherapy (docetaxel, oxaliplatin, leucovorin, and fluorouracil) demonstrated improved survival for locally advanced resectable gastric or gastroesophageal junction adenocarcinoma compared with a regimen similar to the MAGIC protocol. This established FLOT as the preferred chemotherapy-alone regimen.[63]

The Chemoradiotherapy for Oesophageal Cancer Followed by Surgery Study (CROSS) demonstrated the benefits of preoperative chemoradiation (41.4 Gy radiation in 23 fractions with 5 weeks of carboplatin/paclitaxel) and established chemoradiation as the regimen used most often in the United States. Although CROSS included patients with SCC as well as adenocarcinoma of the esophagus and gastroesophageal junction, it confirmed chemoradiation as an effective option for resectable esophageal cancer.[64,65] In 2021, the prospective trial CheckMate 577 found that the addition of adjuvant nivolumab in patients with residual disease following neoadjuvant CROSS chemoradiation improved disease-free survival compared to placebo, and it became the new standard of care at the time.[6]

Recent evidence, however, has raised questions about the necessity of adding radiation to chemotherapy in the neoadjuvant space. Initial results from the multicenter phase III ESOPEC trial demonstrated superior survival when patients with resectable esophagogastric adenocarcinoma were treated with perioperative FLOT as compared with preoperative CROSS-style chemoradiation.[51] Although the possible advantage of chemotherapy alone over chemoradiation was novel, the finding that chemoradiation was not superior to chemotherapy alone was not particularly surprising. Notably, long-term follow-up from CROSS showed a smaller survival benefit for adenocarcinoma than for SCC.[66] Additional studies, including TOPGEAR, NeoRes, and Neo-AEGIS, also found no survival advantage when preoperative chemoradiation was compared to or added to perioperative chemotherapy.[10,67,68] Considering these recent findings, perioperative FLOT chemotherapy should be preferred over CROSS chemoradiation for most patients with locally advanced adenocarcinoma.

Chemoradiation may still be the better option for some patients with locally advanced esophageal adenocarcinoma. The ESOPEC trial results showed that only 68% of patients completed the CROSS induction regimen—far lower than the 87% who tolerated the more intense FLOT regimen, raising concern about optimal delivery of the CROSS regimen. In addition, the ESOPEC trial did not directly compare perioperative FLOT chemotherapy with chemoradiation plus immunotherapy, which became the standard of care after the publication of CheckMate 577 in 2021. For now, treatment decisions should be individualized on the basis of patient fitness and tolerance. Neoadjuvant chemoradiotherapy (nCRT) may be preferred for borderline surgical candidates, for whom radiation may be the primary local control intervention, or patients unlikely to tolerate FLOT. At the time of this publication, despite an approval for adjuvant immunotherapy after chemoradiotherapy per CheckMate 577, there remains no approval for perioperative immunotherapy in conjunction with chemotherapy. However, approval for adding durvalumab to perioperative FLOT is now expected as the phase III MATTERHORN trial achieved its primary event-free survival end point, with data presentation pending.[69] Surgeons must be aware that the landscape of preoperative and perioperative therapy will continue to evolve as further results from the MATTERHORN trial and other studies become available.

THOR

2.3. Definitive chemoradiotherapy (dCRT) for locally advanced esophageal adenocarcinoma is reasonable to consider in patients who are not surgical candidates. CoR: IIa LoE: B-NR

We recommend concurrent chemoradiation as the nonoperative option most capable of achieving long-term control for adenocarcinoma patients when surgery is not planned or is unlikely to be possible. Although the outcome is superior when multidisciplinary treatment of adenocarcinoma includes esophagectomy, surgery may not be utilized for some patients because the tumor is not resectable, the patient is not an appropriate medical candidate, or as a consequence of strong patient preference. There are randomized data demonstrating reasonable survival outcomes with dCRT in nonsurgical candidates in trials investigating chemoradiotherapy against chemoradiotherapy with other therapies. For example, the standard dCRT control arm of the prospective multi-institutional trial RTOG 0436 demonstrated a median overall survival of 19 months with a 2-year overall survival of 44.0% (95% confidence interval [CI], 36%-51%).[70] The 36-month rate of local treatment failure in this trial, including patients on the control arm and those who also received added cetuximab, which was not beneficial, was 48.8% (95% CI, 40.8%-57.5%) and 48.8% (95% CI, 40.8%-57.5%), respectively. Patients who had a clinical complete response (cCR; 53% of patients as determined by endoscopy with biopsy of suspicious areas) had substantially better survival (hazard ratio [HR], 0.46), with a 3-year survival of 43.3% (35.6%-50.6%). Similarly, in the multi-institutional prospective SCOPE-1 trial, the median overall survival for patients with adenocarcinoma was 25.8 months (95% CI, 12.5-46.6; n = 32; 3-year overall survival, 43.8% (95% CI, 26.5-59.8%).[71] The median time to local failure was 27.9 months (95% CI, 19.2-51). These results are notably superior to older trials reporting results of small groups of adenocarcinoma patients such as RTOG 85-01, which reported a 3-year survival rate of 17% (CI, 2-33%) and 5-year survival rate of 13% (CI, 0-27%) and which generated skepticism about the value of definitive radiation in this population.[72] The optimal dose of radiation for definitive management when administered with concurrent chemotherapy remains uncertain, but doses greater than 50 to 50.4 cGy have not been shown to be beneficial. In addition, doses of 41.4 to 50.4 cGy appear to be appropriate on the basis of prospective trials RTOG0436 and SCOPE-1. There is not sufficient evidence to support systemic therapy alone as a definitive therapy for locally advanced disease. The impact of adding immunotherapy is under investigation in the KEYNOTE-975 trial.[73]

2.4. For patients deemed to have a cCR after neoadjuvant chemoradiation, it is reasonable to proceed with esophagectomy, as radiographic and endoscopic surveillance have inadequate sensitivity to reliably identify persistent/recurrent disease. CoR: IIa LoE: B-randomized (R)

Approximately one-third of patients undergoing nCRT have a cCR by radiographic findings and endoscopic biopsy. The recent SANO trial, a randomized phase 3 stepped-wedge cluster trial that evaluated overall survival in patients enrolled in active surveillance versus standard esophagectomy after chemoradiotherapy, found that overall survival from the day of cCR at 34 months in the active surveillance group was noninferior compared with surgery (HR, 0.88 [95% upper boundary, 1.40]; $P = .007$) using a 15% noninferiority margin.[74] Overall, 35% of patients maintained cCR, 48% developed locoregional regrowth, and 17% developed distant metastases. Median disease-free survival was 35 months (95% CI, 31-41) for active surveillance and 49 months for standard surgery (95% CI, 38-NR). At 30 months after nCRT, 43% of patients undergoing active surveillance developed distant metastases versus 34% in the surgery arm.

These data do have limitations. Despite active surveillance showing noninferiority to surgery in terms of overall survival, a significant portion of patients experienced local recurrence, disease progression, and higher rates of distant metastases. Both patients with SCC and adenocarcinoma were included in the per-protocol analysis, as well as nonrandomized patients from the previous pre-SANO study. In addition, there was minimal surgical oversight. There was a very high rate of postsurgical complications in both the active surveillance and surgical groups (82% and 84%, respectively) as well as a high rate of anastomotic leak (22% and 27%), which limits the conclusions of the trial.

In addition, data suggest that a delayed esophagectomy may produce inferior oncologic and clinical outcomes.[75] For example, NeoRes II was a prospective randomized trial that compared surgical intervention within 4 to 6 weeks of completion of nCRT with 10 to 12 weeks of nCRT. The results were negative; however, subgroup analysis demonstrated that patients with worse tumor regression grade have inferior survival when undergoing observation.[76] Concerns exist regarding the limited accuracy of current imaging and endoscopic modalities in determining cCR after neoadjuvant therapy in patients with esophageal cancer. Existing modalities such as CT, PET-CT, and EUS seem inadequate in identifying complete responders accurately. A meta-analysis involving 3625 patients from 56 studies indicated varying sensitivities and specificities of these modalities in detecting complete response.[77] Although multimodal restaging seems to enhance the detection of complete response, the reliance on such methods alone may not suffice to forgo esophagectomy at present. Research on new biomarkers, such as circulating tumor DNA (ctDNA), might enhance tumor progression detection during active surveillance, but further studies are warranted.[78,79] Improved diagnostic tests are necessary to enhance restaging accuracy for individuals with esophageal cancer, with esophagectomy after neoadjuvant therapy still being preferred when medically feasible because of the risks of progression and metastasis during active surveillance.

2.5. If surgery is deferred after chemoradiation for esophageal adeno carcinoma, surveillance should be done by endoscopy ± EUS, as well as imaging (PET-CT strongly preferred), every 3 to 4 months for at least the first 2 years and a minimum of annually thereafter for a total of 5 years. CoR: I LoE: C-LD

In patients with a cCR after neoadjuvant therapy, the surveillance-only approach with salvage surgery in the event of local recurrence is debated, with some limited data demonstrating worse survival in these scenarios.[80-83] The preSANO trial was a prospective phase 2 study investigating the effectiveness of the evaluation of cCR on the basis of tumor regression seen in surgical specimens. They found that, beginning 10 to 12 weeks after the completion of nCRT, evaluation with EUS, bite-on-bite biopsy, and fine-needle aspiration of suspicious lymph nodes is adequate for the detection of residual disease, with PET-CT used for detection of metastases.[84] The subsequent prospective randomized SANO trial, a phase 3 stepped-wedge cluster trial of patients with esophagogastric cancer randomized to surgery or active surveillance, found noninferior overall survival of the active surveillance group compared to standard of care, with a noninferiority margin of 15%.[74] However, these results should be interpreted with caution. Survival data by histologic subtype are not available, and it is known that SCC responds better to nCRT than adenocarcinoma.[64] As discussed above in statement 2.2, data suggest salvage esophagectomy produces inferior oncologic and clinical outcomes.[63]

NeoRes II, a prospective randomized trial that compared surgical intervention within 4-6 weeks of completion of nCRT with surgical intervention 10 to 12 weeks of nCRT, did not demonstrate a significantly greater risk of death during follow-up for patients who waited to have surgery.[76] However, subgroup analysis demonstrated worse survival, specifically for those patients with a worse tumor regression grade who underwent delayed surgery. Lastly, radiographic and endoscopic evidence of complete response are not perfect biomarkers of disease.[85]

If the decision is made to defer esophagectomy, either in the case of careful consideration of the risks of surveillance or if a patient is unfit for an operation, then that patient requires intensive surveillance. Most evidence for surveillance in these cases comes from studies of patients who have undergone nCRT and esophagectomy. It is known that the majority of recurrences after esophagectomy occur in the first 2 years and that

many patients present with symptoms even while undergoing surveillance.[86] To mitigate this risk, in the preSANO trial, patients undergo fluorodeoxyglucose-PET/CT, upper endoscopy with biopsy, and EUS–fine-needle aspiration of suspicious lymph nodes every 3 months for the first year, every 4 months in the second year, every 6 months in the third year, and yearly thereafter for 5 total years.[85] Using this approach, recurrent locoregional disease or new metastases should be identified in a timely manner. Again, we recognize that this frequency of PET imaging may not be feasible for all patients due to insurance coverage or cost. However, each surveillance technology is not perfect in isolation, and the inclusion of PET in the surveillance strategy along with plain cross-sectional imaging for these patients is critical to identify early distant recurrence.

2.6. We recommend testing all patients for mismatch repair at diagnosis. CoR: I LoE: B-R

Mismatch repair-deficient/microsatellite instability-high (MSI-H) tumors are known to have high tumor mutation burdens and excellent pan-tumor ICI efficacy.[87] Although ICI had previously been relegated to metastatic therapy, it is now also the recommended therapy for patients with locally advanced MSI-H tumors.[88-91] Conversely, retrospective evaluation of MAGIC and CLASSIC suggest that perioperative/adjuvant chemotherapy alone may actually be detrimental for patients with MSI-H locally advanced esophagogastric cancers, rather than surgery alone, and so pretreatment mismatch repair testing is crucial, regardless of stage.[92,93]

Although mismatch repair immunohistochemical testing is recommended due to cost, rapidity, and availability, alternatives include microsatellite instability testing by tissue or plasma ctDNA next-generation sequencing. Of note, 7% of results may demonstrate discordance between protein and DNA testing approaches, and the absence of MSI-H by ctDNA should be viewed cautiously in the setting of low plasma tumor DNA shedding.[94,95]

2.7. For mismatch repair-deficient patients, we advise discussion of upfront ICIs and/or surgery. CoR: IIA LoE: B-NR

The GERCOR-NEONIPIGA study identified a pathologic complete response in 17 of 29 patients (58.6%) who received 3 months of neoadjuvant nivolumab and ipilimumab followed by 9 months of adjuvant nivolumab. Similarly, the INFINITY trial achieved a pathologic complete response in 9 of 15 patients (60%) treated with durvalumab and tremelimumab for 3 months. These findings suggest that approximately 60% of patients with MSI-H tumors may not require surgical resection, although data on how to properly conduct surveillance after definitive ICI, as now recommended in the National Comprehensive Cancer Network guidelines, are lacking. INFINITY suggested a high failure rate (5/6 patients) in patients with T4 tumors, so multidisciplinary case review remains crucial for identifying patients who may still benefit from primary tumor resection. It also remains unclear whether therapy failed because of inadequate ICI duration or if CTLA-4 inhibition or concurrent chemotherapy is beneficial. It is also important to note that 19% of patients receiving ICI had grade 3/4 toxicities in NEONIPIGA and INFINITY that may have lifelong consequences. For patients with contraindications to ICI or concerns regarding their ability to undergo adequate surveillance after definitive ICI, surgery remains an effective and highly curative option in this population, with a historically favorable prognosis.

2.8. For patients with extensive nodal disease and/or "nonregional" lymph nodes, upfront systemic therapy may be preferred. CoR: IIb LoE: C-LD

Extent and location of nodal involvement correlate with the likelihood of disseminated disease, and, therefore, are important determinants of survival—particularly in esophageal adenocarcinoma, when compared with the more radiosensitive squamous cell esophageal cancers.[96] In fact, the survival of patients with IVA disease (largely, patients with N2 and N3 disease) mirrors the survival of patients with stage IVB disease.[97,98]

For the purposes of this discussion, extensive nodal disease will be considered N2 (3-6 regional lymph does) or N3 (≥7 regional lymph nodes) disease. Disease in nonregional lymph nodes is considered M1 and, thus, stage IVB disease. Relevant lymph node stations are detailed in Figure 2.[48]

Given the high risk of disseminated disease in patients with extensive nodal disease, upfront systemic therapy to treat micrometastatic disease is preferred. In fit patients, neoadjuvant systemic therapy with FLOT is the current standard of care, likely soon in combination with durvalumab, once the MATTERHORN results are reviewed by the FDA.[51,63,99,100] This paradigm is further supported by recent findings from ESOPEC, where perioperative FLOT led to superior 3-year overall survival rates when compared to chemoradiotherapy (HR, 0.70; 95% CI, 0.53-0.92; $P = .012$).[51]

2.9. Patients with oligometastatic disease at diagnosis should undergo systemic therapy before being considered for locoregional therapy. CoR: IIa LoE: B-NR

Although metastatic esophagogastric adenocarcinoma prognosis remains quite poor, approximately 10% of patients with M1 disease have long-term survival.[96] This survival curve likely represents patients with oligometastatic disease, defined as having 1-5 treatable solid organ metastatic lesions.[101] In these patients, locoregional therapy may be a reasonable approach after demonstrating favorable biology and achieving maximal disease control on systemic therapy previously.[98]

This approach was evaluated in the phase 2 AIO-FLOT3 trial, whereby patients with resectable or metastatic gastric or gastroesophageal adenocarcinoma were stratified into resectable, limited metastatic, and extensive metastatic disease. Patients received 4 cycles of preoperative FLOT followed by surgery and 4 cycles of postoperative FLOT. In the cohort with limited metastatic disease, median overall survival was 22.9 months, with a response rate of 60%, which was increased in comparison to those patients with disseminated disease, with an OS of 10.7 months (HR, 0.37; 95% CI, 0.25-0.55; $P < .001$).[102]

However, the phase 3 RENAISSANCE trial, which randomized patients into 2 months of chemotherapy followed by resection of the primary tumor and distant disease versus continuation of systemic therapy, failed to demonstrate a survival benefit with the addition of surgery. Of note, as seen in FLOT3, while patients with liver and peritoneal metastases did not benefit from this approach, there was a suggestion that patients with retroperitoneal lymph nodes-only distant disease did benefit from resection.[103] This finding is supported by a retrospective review by Tankel and colleagues,[104] which demonstrated a survival benefit to retroperitoneal lymph node dissection in addition to esophagectomy after a prolonged course of systemic therapy.[104]

In the aforementioned prospective studies, restaging was performed after 2 months of systemic therapy or prior to consideration of locoregional treatment. Although the prospective studies above used 2 to 4 months of systemic therapy before locoregional treatment, many retrospective studies use at least 6 months of systemic therapy prior to consideration of locoregional treatment.[105-108] In light of available data, at least 6 months of systemic therapy is recommended before consideration of locoregional therapy to the primary and metastatic sites in a patient with oligometastatic disease. Interim restaging scans for response assessment should be performed every 2 to 3 months, with a multidisciplinary case review before surgical consideration. In addition, restaging diagnostic laparoscopy should be considered in these patients before consideration of surgery or other local therapy to the primary tumor or oligometastases. Although neither the RENAISSANCE or FLOT3 trial directly addresses restaging laparoscopy, it is important to rule out peritoneal or otherwise untreatable disease in these patients following systemic therapy.

Importantly, the role of local therapy in oligometastatic disease remains highly controversial, and thorough multidisciplinary case review is critical for these patients. We do not recommend the routine implementation of local therapy in patients with oligometastasis. The data discussed in this

section apply only to a very highly selected cohort of patients. Any consideration of locoregional therapy in these patients is a high-risk endeavor and should take place only in high-volume centers with multidisciplinary support.

Section 3: SCC

3.1. For patients with SCC, definitive endoscopic therapy is recommended only for those patients with superficial lesions with a low risk of lymph node involvement. CoR: I LoE: B-NR

Early-stage SCC is overall more aggressive than adenocarcinoma, and, therefore, a specific approach should be used when treating this disease. Superficial SCC can be treated with definitive endoscopic therapy if the lesion has a low risk of lymph node involvement. Those who are low risk have lesions that are limited to the mucosal epithelium, mucosal lamina propria, or muscularis mucosa. These superficial tumors have a low risk of lymph node involvement, particularly if the lesions have an absence of LVI, a length <2 cm, and the absence of fluorodeoxyglucose-avid lymph nodes on PET-CT examination before treatment.[109-113]

The risk of lymph node involvement dramatically increases with increasing depth of tumor invasion. In a large database study, T1a tumors had a 5% risk of lymph node involvement, and T1b tumors had a risk of almost 17%.[114]

Beyond the depth of invasion, LVI is an extremely important risk factor. In a large series of EMR for T1a tumors subdivided into M1 (intraepithelial), M2 (tumors invading the lamina propria), and M3 (tumors in contact with or invading the muscularis mucosa), there was significant lymph node involvement (42%) for M3 tumors with LVI.[114] Further, another study of T1a SCC treated with EMR found a 5-year metastasis rate of 46.7% versus 0.7% for those with and without LVI ($P < .0001$).[115] Low-risk patients can be considered for surgical resection on the basis of patient preference and a multidisciplinary discussion. The benefit of surgical resection is more accurate staging, as even low-risk superficial lesions can have lymph node involvement.[44] Patients found to have unsuspected lymph node involvement will benefit from adjuvant therapy.[116]

High-risk patients are those with LVI and total depth of invasion beyond the muscularis mucosae. These patients should more strongly consider esophagectomy at the time of diagnosis due to the poor prognosis of untreated lymph node disease.

3.2. For patients with high-risk SCC who undergo definitive endoscopic therapy only, we suggest maintaining a strict follow-up schedule to detect any recurrent disease at a curable stage, by endoscopy ± EUS every 3 months for the first year, every 6 months during the second year, and a minimum of annually thereafter. CT (prefer PET) is advised every 6 months for 2 years and then annually for 5 total years. CoR: IIa LoE: C-LD

After endoscopic resection of esophageal SCC, patients are at increased risk of occult lymph node metastasis and metachronous esophageal cancer. High-risk patients are those with an increased risk of lymph node involvement, such as those with LVI and a depth of tumor invasion beyond the muscularis mucosae. Both are found on the pathology report following endoscopic resection. Other high-risk features include positive margins, tumors larger than 2 cm, poor differentiation, and submucosal layer invasion exceeding 200 μm.[110,117-119]

On a meta-analysis of nearly 2000 patients with T1a and T1b esophageal SCC, lymph node involvement was more significantly present in tumors >2 cm in size, tumors located in the lower esophagus, tumors invading the submucosa (T1b), poor differentiation, and LVI.[120] In numerous studies, LVI has been found to be a strong independent risk factor for lymph node involvement and worse overall survival.[120]

Esophagectomy is the preferred approach for patients with high-risk SCC. High-risk patients should be counseled on the benefits of surgical resection. These high-risk patients who opt out of surgery should undergo intensive surveillance to identify recurrence early and improve long-term survival. Currently, there are limited studies examining the impact of different surveillance strategies on the early detection of recurrence and the impact on mortality after endoscopic resection for esophageal SCC. Given the limited data, recommendations are based on studies primarily aimed at adenocarcinoma and head/neck SCC. These studies confirm that the most effective surveillance is with endoscopy ± EUS every 3 months for the first year, every 6 months during the second year, and a minimum of annually thereafter, as well as PET scan every 6 months for 2 years and then annually for 5 total years.[80] This surveillance protocol is supported by previous studies showing that the majority of recurrences occur within the first 2 years after surgical resection of esophageal cancer.[86] Consideration can be given to continued surveillance beyond 5 years, depending on patient risk factors. In addition, although we recognize that this frequency of PET imaging may not be feasible for all patients due to insurance coverage or cost, the inclusion of PET as a component of the surveillance strategy is critical to identify recurrence.

There is emerging data on the combination of adjuvant therapy with either chemotherapy and radiation or radiation alone following endoscopic resection for esophageal SCC when esophagectomy is omitted.[121,122] The theory is to prevent locoregional recurrence in patients deemed as high risk. There is no standard adjuvant approach at this time, but continuing studies are looking at patients who will gain the most benefit from the combination of endoscopic resection with adjuvant therapy.

3.3. Surgery should be considered in patients with esophageal SCC with a high risk of tumor recurrence despite a cCR to chemoradiation. CoR: I LoE: B-NR

Patients at high risk of tumor recurrence include those in whom locoregional residual tumor is highly suspected or proven.[84] Additional factors that may increase the risk of tumor recurrence, despite a cCR after chemoradiation, are poor differentiation or high-grade tumor and the presence of LVI.[123,124] In the presence of any of these high-risk factors, curative-intent surgery is recommended.

Presently, studies on organ preservation for esophageal cancer involving both early stages and advanced stages are mixed. The JCOG0502 study included only clinical stage I esophageal SCC and demonstrated a cCR of 87% after curative chemoradiotherapy, with comparable long-term survival between chemoradiotherapy alone and surgery alone.[125] In addition, the JCOG9906 and JCOG0909 studies evaluated the use of dCRT with the option of salvage resection in clinical stage II/III (no T4) esophageal SCC patients.[126,127] These trials reported a cCR up to 65.0% with a 5-year overall survival of up to 36.8%. Notably, the 5-year overall survival for patients requiring salvage esophagectomy was only 22%. Lastly, the JCOG9516 study included clinical stage T4 patients who demonstrated a cCR of 15% and a poor median survival of <1 year.[128]

Although a considerable number of patients in the above studies had cCR and long-term survival after chemoradiotherapy, approximately one third of patients experienced local recurrence.[126,127] Patients with recurrence have limited systemic options but are candidates for salvage surgery. Salvage esophagectomy is not without risk, and, therefore, curative-intent surgery is recommended for patients with a high risk of recurrence after dCRT.[126-130]

3.4. In patients with recurrent or persistent esophageal SCC after dCRT, salvage esophagectomy is recommended, but it should be performed at experienced centers due to the greater risk of complications. CoR: I LoE: B-NR

dCRT without planned surgery is a treatment option for SCC of the esophagus.[1] We addressed the uncertainty about whether patients treated with dCRT who have a cCR should be considered for planned esophagectomy or whether it would be more appropriate to wait for progression. There was strong consensus that surgery should proceed when there is

persistent local disease in the absence of metastatic failure, confirmed by restaging with the goal of achieving long-term survival and preventing the symptom burden of locally progressive disease. Given the evidence for increased surgical risk after chemoradiation in this population, we recommend that this procedure and postoperative care be performed by experienced care teams.

The approach of dCRT for SCC is supported by the higher pathologic complete response observed in SCC compared to adenocarcinoma (49% vs 23%) as well as 2 phase 3 trials accruing patients primarily in the late 1990s that did not show a survival benefit to planned esophagectomy added to chemoradiation.[64] Nevertheless, 40% to 60% of patients will not have a cCR and/or will have locally recurrent disease during their lifetime.

In a prospective study with planned salvage surgery for patients with SCC who do not have a cCR, long-term disease control with 3- and 5-year survival of 34% and 22%, respectively, was achieved.[127] An additional prospective trial, FFCD 9102, demonstrated a 3-year survival of 30% in patients with SCC who received an esophagectomy after inadequate response to chemoradiation.[131] On the basis of these and other studies, we recommend esophagectomy when persistent or recurrent local disease persists after neoadjuvant therapy, rather than waiting for progression. Unaddressed locally persistent disease increases the risk of avoidable metastatic failure, significant dysphagia, and progressive fibrosis from radiation over time. There is no validated approach to clinically confirm a high likelihood of complete response. We recommend a comprehensive assessment that includes PET scan and endoscopy (with or without ultrasound) and biopsy as indicated to inform optimal response-based decision-making.[132]

Surgery after dCRT has a high risk of postoperative morbidity and mortality. In one single-institution review of patients undergoing salvage esophagectomy following failure of dCRT for esophageal SCC, 90-day mortality was >17%, with 71% of patients experiencing a major postoperative complication.[133] Keys to improving these outcomes include a comprehensive preoperative risk assessment, a review of the previous radiation plan with anastomoses above the radiated field, consideration of alternative conduits, and a multidisciplinary discussion.[134]

3.5. If surgery is deferred after chemoradiation for esophageal SCC, surveillance should be done by endoscopy ± EUS, as well as imaging (PET-CT strongly preferred), every 3 to 4 months for at least the first 2 years and a minimum of annually thereafter for a total of 5 years. CoR: I LoE: C-Expert Opinion

Although a fairly uncommon scenario, some operative candidates with locally advanced, resectable disease may be offered a treatment strategy whereby confirmation of persistent or late recurrent local disease after neoadjuvant treatment (chemoradiation) is required before esophagectomy.[135] The goal in this circumstance is organ preservation; this can become confounded by a potential delay in treatment, thereby permitting disease escape and subsequent cancer-related mortality. Consequently, the intensity of follow-up needs to be escalated.

For a patient to be offered this alternative curative approach, it would be assumed that there was locally advanced disease characterized by standard modalities (esophagogastroduodenoscopy/EUS, PET-CT) and that the intention to treat would include chemoradiotherapy and esophagectomy. Most patients subject to this recommendation would likely have been treated with CROSS therapy. Regardless, all individuals would be mandated to have had a cCR and presumably remain candidates for esophagectomy. Those suffering toxicity from neoadjuvant therapy or those with an incomplete clinical response would fall out of the purview of this recommendation.

The sensitivity of fluorodeoxyglucose-PET for esophageal SCC recurrence after curative intent treatment approaches 100%.[136] Routine esophagogastroduodenoscopy/EUS and a bite-on-bite approach should be adopted whereby biopsy sites positive for cancer at the beginning of treatment are sampled again at each surveillance episode.[84]

Surveillance intensity should be highest in the first 2 years after chemoradiation, with follow-ups every 3 to 4 months, as used in trials involving this unique patient population.[84,135] Given that the risk of recurrence declines significantly beyond 2 years in patients with locally advanced disease treated with curative intent, de-escalation of surveillance may be considered at that point.[137] We recognize that this frequency of PET imaging may not be feasible for all patients due to insurance coverage or cost. However, the inclusion of PET in the surveillance strategy along with plain cross-sectional imaging for these patients is critical to identify recurrence.

In the future, additional interventions to assist esophagus-preservation strategies in this setting might include an assay of ctDNA, as this technology has shown promise to become an important predictor of recurrence after chemoradiation.[79]

3.6. SCC of the cervical esophagus should be treated with chemotherapy and radiation, with surgery reserved only for select patients. CoR: I LOE: B-NR

The location of the tumor in SCC of the esophagus (cervical, mid-, or lower/gastroesophageal junction) is relevant when considering surgery as part of definitive management. SCC of the cervical esophagus (defined as the upper esophageal sphincter to the thoracic inlet) is often locally advanced at the time of diagnosis. Because of the proximity of the cervical esophagus to vital structures such as the larynx, upper airway, and spinal cord, nonsurgical management is the current preferred therapeutic option. This contrasts with SCC of the mid- and lower esophagus, where surgical resection is more standard. Surgical resection for cervical esophageal SCC is a highly morbid operation and is associated with an overall poor quality of life. Often, a resection of the larynx and/or pharynx is required. The alternative to surgical resection with similar overall survival is dCRT.[138,139] This approach is now part of the American Society of Clinical Oncology guidelines and the National Comprehensive Cancer Network guidelines. Both recommend dCRT for tumors in the cervical esophagus, with salvage surgery reserved for persistent or recurrent disease.[140] Salvage surgery is associated with significant morbidity and mortality.[75,133] It should only be offered as part of a multidisciplinary discussion and preferably at a center that does a large volume of esophageal resections.

In regards to the radiation approach, intensity-modulated radiation therapy is preferred over 3D conformal radiotherapy for cervical esophageal carcinoma because normal radiosensitive structures in the head and neck are close to the radiation field.[141] Radiation doses are commonly 50.4 Gy but can be taken up to 70 Gy in the cervical esophagus based on head and neck literature.[142] These greater doses have a rate of serious adverse events as high as 10%, including dysphagia, strictures, fistulas, dehydration, mucositis, esophagitis, dermatitis, and fatigue, all without the guarantee of an increased tumor response.[72,143]

Despite the introduction of dCRT, survival rates remain relatively low, with an overall poor quality of life. There is currently no consensus on the role of immunotherapy in these patients. There are ongoing studies looking at the role of induction PD-1 inhibitors in combination with chemotherapy prior to radiation.[143] For example, the KN975 trial is examining the role of first-line pembrolizumab following dCRT in patients with unresectable esophageal cancer, including both adenocarcinoma and SCC.[73] The role of immunotherapy in SCC will become evident with more ongoing clinical trials.

DISCUSSION

The topics covered in this document were developed on the basis of literature search results and the expertise of the authors. It should be noted that randomized data that support esophageal cancer management are somewhat limited, with a subsequent impact on the level of evidence used to support the group's recommendations.

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The prognosis for esophageal and gastroesophageal junction cancer has improved significantly over the past few decades, likely the result of a combination of discovering more cancers at an earlier stage, increased use of multimodality therapy that includes esophagectomy, and improved short-term esophagectomy outcomes. Although esophagectomy has historically been considered necessary to provide a patient with esophageal cancer with a cure, its role in both early-stage superficial cancers and locally advanced cancers has evolved. In the future, there may be a role for active surveillance in those patients who appear to have a cCR after systemic therapy, per the SANO trial. However, more data is needed before embracing this approach.[74] Our current radiographic and histologic surveillance technologies are imperfect, and we caution against routine use of the active surveillance strategy in the context of current standard-of-care guidelines. There are exciting new technologies in esophageal cancer therapy, including the development of improved biomarkers and liquid biopsy in the form of ctDNA, that may improve our ability to appropriately select patients for surgery and to improve our surveillance accuracy.[79]

The availability of endoscopic resection for early-stage disease, as well as increased evidence regarding the benefits of induction chemoradiation and perioperative systemic therapy for more advanced cancers, has revolutionized the field.[4-10] Although the morbidity of esophagectomy has decreased with the advent of minimally invasive approaches, high-volume centers, at-home symptom monitoring, and enhanced recovery after surgery programs, this operation still has significant potential to negatively impact the quality of life of the patient.[144] The choice to operate on these patients should be made after careful consideration of the evidence, multidisciplinary review, and thorough discussion with the patient regarding recovery.

Our work provides a basic framework for the approach to most patients with potentially resectable esophageal cancer, although data will continue to evolve as more trials become available to review.[51,99] The recent publication of the ESOPEC trial has led to neoadjuvant FLOT chemotherapy largely replacing CROSS-era chemoradiation plus adjuvant immunotherapy as the standard of care for most patients.[51] However, as outlined in this document, nCRT remains appropriate in select clinical situations, underscoring the importance of multidisciplinary discussion in these complex cases.

In addition, the use of induction ICI in the care of patients with esophageal cancer is under investigation.[100] Although several prospective studies have demonstrated a benefit of induction ICI in MSI-high tumors, its use in combination with chemotherapy is still being evaluated. At the time of publication of this document, a press release reveals that the addition of durvalumab to perioperative FLOT in the phase 3 MATTERHORN trial has achieved its primary event-free survival end point. The upcoming publication of the full MATTERHORN trial results may again change the standard of care for resectable esophagogastric cancer patients to include FLOT plus immunotherapy.[69,99]

We hope this document serves as a valuable resource for clinicians navigating the evolving and complex landscape of esophageal cancer management. As new therapies and diagnostic tools emerge, patient centered, personalized care has become increasingly relevant.[6,145-147] Patient-reported outcomes, technology to track at-home recovery, and tools to support the psychosocial wellness of our cancer patients have become more commonplace.[144,148-150] As therapy options continue to expand, the need for access to specialized care, expert endoscopists, esophageal surgeons, gastrointestinal pathologists, and specialized medical and radiation oncologists at high-volume centers will continue to become more important.

Webcast

You can watch a Webcast of this AATS meeting presentation by going to: https://www.jtcvs.org/article/S0022-5223(25)00298-3/fulltext.

Conflict of Interest Statement

Dr Enzinger is a consultant for and received honoraria from Astellas Pharma, BeiGene USA, Daiichi Sankyo, Eisai, Legend Biotech, Merck & Co, Novartis, Oncolys BioPharma, Regeneron, Sanofi-Aventis, and Servier Pharmaceuticals. Dr Kleinberg received grants or contracts, consulting fees, and payment or honoraria for lectures, presentations, speakers bureaus, manuscript writing or educational events from Novocure (institution), Novartis (institution), BMS (institution), Incyte (institution), Accuray (personal), and Servier (personal). Dr Maron received honoraria from Novartis, Amgen, Elevation Oncology, Pinetree Therapeutics, Purple Oncology, Bolt Therapeutics, and Daiichi Sankyo; has a financial interest in OneCellDx; and received research funding from Conquer Cancer Foundation, research travel support from AstraZeneca, and research support from AstraZeneca and Paige.AI. Dr Molena received consulting fees, payment or honoraria for lectures, presentations, speakers bureaus, manuscript writing or educational events, support for attending meetings and/or travel, participation on a Data Safety Monitoring Board or Advisory Board: AstraZeneca, J&J, Boston Scientific, Boehringer, BMS, Ideology health, and Merck. Dr Murthy reports consultant for OnTarget, Lazzaro Medical, Inc, and Arthrex; and speaker for Medtronic and Intuitive Surgical Inc. Dr Palta reports research funding to institution from Merck and Varian Medical Systems; royalties from UpToDate; advisory board of Varian Medical Systems; board of ASCO GI meeting Committee; ASTRO Education and CME Committee. Dr Pouw reports consultancy fee from icroTech Europe, consultancy for Medtronic B.V. (payment to institution); speaker fees from Pentax B.V. and Boston

Scientific; and research grant from Möller Medical (payment to institution). Dr Reddy reports grants or contracts, consulting fees, payment or honoraria for lectures, presentations, speakers bureaus, manuscript writing or educational events, participation on a Data Safety Monitoring Board or Advisory Board, and leadership or fiduciary role in other board, society, committee or advocacy group, paid or unpaid: Intuitive, AtriCure, Genentech, On Target Labs, Medtronic, Bristol Myers Squib, General thoracic surgical club, and MIXR Center. Dr Schumacher reports consulting fees from Intuitive Surgical and Medtronic. Dr Sihag reports consulting fees and payment or honoraria for lectures, presentations, speakers bureaus, manuscript writing or educational events from Astra Zeneca, Intuitive Surgical, and Medtronic. Dr Worrell consulting fees and payment or honoraria for lectures, presentations, speakers bureaus, manuscript writing or educational events from Intuitive and Genentech. All other authors reported no conflicts of interest.

The *Journal* policy requires editors and reviewers to disclose conflicts of interest and to decline handling or reviewing manuscripts for which they may have a conflict of interest. The editors and reviewers of this article have no conflicts of interest.

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Key Words: esophageal cancer, adenocarcinoma, squamous cell carcinoma

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APPENDIX E1. AATS EXPERT CONSENSUS DOCUMENT: EXPECTATIONS AND APPROACHES OF ESOPHAGECTOMY

Search Strategy

We developed searches that incorporated key words and Medical Subject Headings (MeSH) relating to the following concepts: esophageal cancer, esophagectomy, early-stage disease, resection, recurrence, and complications. Searches were limited to human studies published in English since January 1, 2003.

Notes

- The final search strings will find the words listed below and their variants (plural and singular form)
- Asterisk—indicates truncation and tells the database to find all variants of a word. The final search will apply truncation more generously.
- MeSH/Majr are Medical Subject Headings for the MeSH thesaurus.
- We will search the titles and abstracts for the keywords.
- Search concepts are highlighted

Concept: Esophageal Cancer

MeSH

- "Esophageal Neoplasms"[Mesh]
- ("Esophagus"[Mesh/"Esophagus"[Majr]) with ("Neoplasms"[Mesh]/"Neoplasms"[Majr])

Key words:

esophageal oesophageal esophagogastric oesophagogastric gastroesophageal food pipe	Near	Cancer OR cancerous OR carcinoma OR adenocarcinoma OR tumor OR tumour OR neoplasm OR neoplastic OR malignancy OR malignant OR metastasis OR oncology

Concept: Esophagectomy

- "Esophagectomy"[Mesh]
- esophagectomy
- Esophageal surgery
- Surgery
- Surgical

Concept: Early-Stage Disease

Key words

- early stage
- early stages
- early NEAR cancer
- T1 OR T1a OR T1b
- stage one
- stage I
- stage 1
- stage IA
- IA1
- IA2
- IA3
- IB
- early cancer

Concept: Endoscopic resection

- "Endoscopic Mucosal Resection"[Mesh]
- endoscopic resection
- endoscopic mucosal resection (EMR)
- endoscopic full-thickness resection (EFTR)
- Strip biopsy
- Endoscopic excision
- Endoscopic removal
- Endoscopic surgery
- Endoscopic procedure
- Endoluminal resection

- Indications for surgical resection after endoscopic resection
  - **Margin** Status
    - margin
  - **T1b** tumors - surgery for all or selectively based on path
    - "Pathology"[Mesh] OR "pathology" [Subheading] OR patholog*
  - Use of **PET scans and Endoscopic Ultrasound** to Guide decision regarding surgery
    - "Positron-Emission Tomography"[Mesh] OR PET OR "Positron Emission Tomograph*" OR "Endosonography"[Mesh] OR endosonograph* OR Echo-Endoscop* OR Ultrasonic-Endoscop* OR "Endoscopic Ultrasonograph*"
- **Surveillance** strategy if surgery not performed
  - "Sentinel Surveillance"[Mesh] OR Surveillance OR "Watchful Waiting"[Mesh] OR "Expectant Management"
- Role of surgery for repeated recurrences after endoscopic therapy for high-grade dysplasia or metachronous early-stage tumors
  - **recurrences**
    - "Recurrence"[Majr] OR "Neoplasm Recurrence, Local"[Majr] OR Recurre* OR Recrudescence*
  - **high-grade dysplasia or metachronous**
    - high-grade dysplasia* OR HGD OR "Neoplasms, Second Primary"[Majr] OR metachronous* OR "Second Primary"

- Management of **complications** of endoscopic therapy **(strictures/fistulas/perforations/leaks)**
    - complication* OR "Constriction, Pathologic" [Mesh] OR Stenos* OR "Pathological Constriction*" OR "Pathologic Constriction*" OR "Fistula"[Majr] OR fistula* OR "Esophageal Perforation"[Majr] OR perforation* OR leak*

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Class of Recommendation

		Suggested phrases for writing recommendations
I	**Strong recommendation** Benefit >>> risk	• Is recommended • Is indicated
IIa	**Moderate recommendation** Benefit >> risk	• Should be considered • Is reasonable • Can be useful/effective/beneficial
IIb	**Weak recommendation** Benefit ≥ risk	• May/might be reasonable • May/might be considered
III	**Moderate recommendation** No benefit: risk = benefit **Strong recommendation** Harm: risk > benefit	• Is not recommended • Is contraindicated • Should not be performed/administrated

Level of Evidence

A	High-quality evidence from more than one RCT with no concern for high risk of bias, or meta-analyses of high-quality RCTs with no concerns for significant bias.	
B	Randomized (B-R):	Moderate-quality evidence from 1 or more RCT or meta-analysis of RCTs.
	Non-Randomized (B-NR):	Moderate-quality evidence from 1 or more well-designed, well-executed nonrandomized studies, observational studies, or registry studies or meta-analyses or such studies.
C	Limited data (C-LD):	Studies with limitations of design or execution, meta-analyses of such studies, case series, or physiological or mechanistic studies in human subjects.
	Expert opinion (C-EO):	Consensus of expert opinion based on clinical experience.

AATS, American Association fo Thoracic Surgery: EACTS, European Association for Cardio-Thoracic Surgery; ESTS, European Society of Thoracic Surgeons; STS, Society of Thoracic Surgeons: RCT, randomized controlled trial.

FIGURE E1. Adopted definitions of classes of recommendation and levels of evidence.[151] *RCT*, Randomized controlled trial; *R*, randomized; *NR*, nonrandomized; *LD*, limited data; *EO*, expert opinion.